

William Cruz Molina

Cognitive Science · Computational Behavioral Research · Advanced Statistics

contact

Kaohsiung, Taiwan
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languages

Fluent English &
Spanish
Intermediate High
Chinese

programming

R/RStudio, Python,
MATLAB,
Unity, L^AT_EX, VS Code,
Git, Shell Scripting

core skills

Experimental Design,
Advanced Statistical
Modeling,
Behavioral Data
Analysis,
Spatial Cognition & VR
Methods,
Interdisciplinary
Research

Summary

Interdisciplinary Cognitive Scientist and PhD Candidate at NCKU specializing in human experimental design, behavioral data analysis, and spatial cognitive processes. Expertise in developing immersive experimental paradigms in Unity and architecting end-to-end data pipelines in R and Python for advanced statistical analysis and computational behavioral modeling. Proven track record in international, cross-cultural research environments with strong focus on understanding human cognitive mechanisms.

Education

- 2019–2026 **Ph.D. Candidate in Mind Sciences** National Cheng Kung University (NCKU), Tainan
VR-based research on spatial transformation: how interactivity modulates mental rotation ability in viewpoint-contingent virtual environments.
- 2012–2015 **M.Sc. Experimental Psychology** National ChengChi University (NCCU), Taipei
Emphasis on Eye Movement Research and technical apparatus calibration.
- 2006–2011 **B.Sc. Psychology** National Colombian University, Bogotá
Majoring in Psychometrics and Statistical Analyses.
- 2018 / 2015 **Proficiency Certificates** TOEFL / TOCFL
English: 101/120 (High Proficiency). Chinese: Level 3 (Intermediate).

Experience (Part I)

- 08/25–Now **Vision Laboratory, NCKU, Tainan** Research Assistant
Design advanced experimental paradigms in Unity for human spatial cognition and basic vision research. Develop custom behavioral tracking scripts and data pipelines using Python and R for statistical modeling.
- 08/25–04/26 **Live On Language CramSchool, Kaohsiung** Language Instructor
Delivered instructional coursework across diverse age groups. Fostered cross-cultural communication skills
- 02/25–08/25 **Sundex, Kaohsiung** Int. Sales Representative & Customer Support
Coordinated international client operations, requirements analysis, and technical specifications management.
- 09/18–02/24 **Cambridge Language School, Tainan** Spanish and English Teacher
Cross-cultural communication and linguistic training for Spanish and English learners.
- 2015–2024 **Various Academic & Private Educational Institutions, Taiwan** Language Tutor & Online Instructor
Conducted language instruction and student evaluations across Cambridge, AMC Group, and American Eagle School platforms.

Experience (Part II)

- 01/15–09/15 **Trend Micro Research Dept., Taipei** Computational Research Intern
Developed text classification and anomaly detection algorithms for automated spam identification, leveraging computational heuristics and natural language processing concepts.
- 09/12–07/15 **Eye Movement and Reading Laboratory, NCCU, Taipei** Cognitive Research Assistant
Programmed and executed eye-tracking paradigms for human cognitive experiments in MATLAB. Managed participant testing, signal collection, and quantitative behavioral data analysis.
- 11/11–04/12 **National University of Colombian, Bogotá** Psychometrician & Quantitative Analyst
Executed advanced statistical analysis and psychometric modeling on complex behavioral datasets to generate empirical risk profiles.
- 09/11–05/12 **Antonio Nariño University, Bogotá** Data Analyst & Research Assistant
Conducted quantitative data analysis and institutional metrics evaluation for large-scale academic audits.
- 03–08/11 **National University of Colombian, Bogotá** Auditor and HR Assistant
Executed systematic validation protocols and data auditing for national candidate selection processes.

Conferences

- 05/2023 **The International Conference of the Applied Psychology in SDGs & Annual Meetings —TAPA, Taiwan** Oral
Is There a Solution to Procrastination? Course Design and Learning-Effectiveness Evaluation of a Self-Regulated-Learning-Based “Psychology of Procrastination” Course. Statistical analysis and data processing.
- 05/2022 **Vision Sciences Society —VSS** Poster
EEG classification of mTBI patients using Random Forest algorithms.
- 07/2019 **15th Asia-Pacific Conference on Vision —APCV, Osaka** Poster
Visual complexity distribution of words and saccade targeting in Chinese sentence reading.
- 08/2013 **5th Asia Congress of Health Psychology, Daejeon** Poster
CBT on Chronic Fatigue Syndrome: A meta-analysis.
- 05/2011 **5th Latin American Congress of Psychology —ALAPSA, Xalapa** Oral
Psychometric analysis of the Eurotest for mild cognitive impairment screening.

Publications

- in prep. **Manuscript** NCKU, Universitat Jaume I, & University of Cincinnati
Interactivity Modulates Spatial Transformation Ability: Evidence from a Viewpoint-Contingent VR Mental Rotation Task.
- in prep. **Manuscript** NCKU & University of New Mexico
EEG Classification of Visuospatial Working Memory in mTBI: Within-Subject Inflation, Diagnostic Null, and LOSO-CV Bench-marking
- in prep. **Manuscript** NCKU & National ChengChi University
The Effect of Word's Visual Complexity Distribution on Saccade Targeting in Reading Chinese Sentences
- 2014 **Avendaño-Prieto, B.L., Avendaño-Prieto, G., Cruz-Molina, W., & Cárdenas-Avendaño, A.** *Diversitas: Perspectivas en Psicología* 10(1), 13–27
Reference guide for non-expert researchers in multivariate statistics.
- 2014 **Avendaño, B.L., Avendaño, X.A., & Cruz-Molina, W.** *Avances en Medición* 9, 58–72
Adaptation to the Colombian context of the screening test for the evaluation of dementia (EuroTest).

Applications and websites

- 2026 **Scroll Master** Speed Reading App
Speed-reading web app for building faster reading pace and comprehension.
scroll-master.wi-cruz.m.workers.dev
- 2025 **Apex Fintech** Algorithmic Trading
Interface development for financial services. apexfintech.net
- 2024 **Market Shift & FVG** MQL5 Market Indicator and Trading algorithms
MetaTrader 5 indicator for market structure and fair value gap detection.
mql5.com